**DE–8035** **13**

DISTANCE EDUCATION

B.Sc. (Computer Science) DEGREE EXAMINATION, MAY 2014.

DIGITAL COMPUTER FUNDAMENTALS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Convert the following :

   (i) $(6BC)_{16} = (\ \ )_2$

   (ii) $(642)_8 = (\ \ )_{10}$

   (iii) $(5A)_{16} = (\ \ )_{10}$

   (iv) $(11010)_2 = (\ \ )_{10}$.

   (b) Convert the decimals 35, 47 to binary and perform $(35 - 47)_{10}$ using 2's complement binary arithmetic.

2. (a) State and prove the basic axioms of Boolean algebra.

   (b) What are sum of products form and product of sums form of logic expressions? Explain with suitable example.

3. (a) Using tabulation method simplify the following Boolean function

   $F(X_1, X_2, X_3, X_4, X_5, X_6, X_7) = \Sigma(28, 20, 60, 52)$.

   (b) Explain the operation of multiplexer with a neat diagram.

4. (a) Explain the construction of ALU in detail.

   (b) What is a full adder? How is a full-adder built using half-adder? Write the truth table for full adder.

5. (a) Explain the working of a parallel binary adder.

   (b) Explain the working principle of an encoder and draw the circuit diagram.

6. (a) Draw the schematic diagram of a master slave JK flipflop. Discuss its working principle.

   (b) What is shift register? What are its various types? Explain its function.

7. (a) What is binary counter? Construct a binary counter using RS flip-flop.

   (b) Discuss the important features of various generations of computers. Give example of computers for each generation.

8. (a) What is the function of memory in a computer? Explain the different types of memory.

   (b) What is an output device? What are the major output devices for a computer? Explain.

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**DE–8036** **14**

DISTANCE EDUCATION

B.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2014.

DATA STRUCTURES USING 'C'

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

($5 \times 20 = 100$)

1. (a) Describe the basic structure of 'C' program. (6)

   (b) What is a string constant? How is the string constant different from a characters constant? Give an example. (6)

   (c) What is the precedence? Explain the order of evaluation. (8)

2. (a) Why do you need the function? Describe the function declaration and accessing a function. (10)

   (b) Explain the multidimensional array with an example. (5)

   (c) Can an array name be used as an argument to a function? Write a program example. (5)

3. (a) Write a short note on pointers and passing pointers to a function. (8)

   (b) What is the difference between the arrays of pointers and pointers to the array? (6)

   (c) Explain how to create data file. (6)

4. (a) Write a 'C' program to search a number in an array. (10)

   (b) Describe the operations that can be performed on stack Implement stack using an array in program. (10)

5. (a) Write a recursive function to calculate factorial of a numbers and a main program to calculate NCR value. (10)

   (b) Narrate linear search method with an example. (10)

6. (a) Write a procedure to evaluate post fix and infix expression. (8)

   (b) What are the three categories of functions and explain with examples? (6)

   (c) What are the advantages of pointers? (6)

7. (a) Explain all type of loop construct in 'C' language with an example. (8)

   (b) Discuss the "Else if" ladder structure in 'C' with an example. (6)

   (c) Explain the concept of Nesting of functions. (6)

8. (a) Explain how elements of a two dimensional array can be accessed using pointers. (10)

   (b) Explain in detail about doubly linked list. (10)

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**DE–8037** **15**

DISTANCE EDUCATION

B.Sc. DEGREE EXAMINATION, MAY 2014.

Computer Science

DISCRETE MATHEMATICS

(2007 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Show that $(\neg p \wedge (\neg Q \wedge R)) \vee (Q \wedge R) \vee (p \wedge R) \Leftrightarrow R$. (7)

   (b) Show that $(P \wedge Q) \vee (\neg p \wedge Q) \vee (P \wedge \neg Q) \vee (\neg p \wedge \neg Q)$ is a Tautology. (7)

   (c) Construct the Truth table for the formula $(p \wedge (p \leftrightarrow q)) \rightarrow q$. (6)

2. (a) Show that $A \cap (B - C) = (A \cap B) - (A \cap C)$. (10)

   (b) Let $A = \{a, b\}$, let $R = \{(a, b), (b, a), (b, b)\}$ and $S = \{(a, a), (b, a), (b, b)\}$ be relations on A. Find $S \circ R$ and $R \circ S$. Comment on your result. (10)

3. (a) Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be both one-to-one and onto functions. Prove that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$. (10)

   (b) Show that the function $f\langle x, y\rangle = x + y$ is primitive recursive function. Hence compute the value of $f\langle 2, 3\rangle$. (10)

4. (a) Define

(i) Semi group

(ii) Monoid

(iii) Group

(iv) Normal subgroup

(v) Group homomorphism. (10)

(b) State and prove Lagranges theorem. (10)

5. (a) Prove that every subgroup of a cyclic group is cyclic. (10)

(b) Prove that a nonempty subset $H$ of a group $(G, *)$ will be a subgroup of $G$ if and only if $a * b^{-1} \in H$ for all $a, b \in H$. (10)

6. (a) Define

(i) Digraph

(ii) Regular graph.

(iii) Complete graph

(iv) Simple graph

(v) Connected graph. (10)

(b) Prove that a tree with $n$ vertices has $(n-1)$ edges. (10)

7. (a) Prove that the graph $K_{3,3}$ is not planar. (10)

(b) Prove that a given connected graph $G$ is Eulerian if and only if all the vertices of $G$ are of even degree. (10)

8. (a) Show that $(\forall x)\ (p(x) \wedge Q(x)) \leftrightarrow ((\forall x))(p(x) \wedge (\forall x)\ (Q(x))$ is a logically valid statement. (10)

(b) Prove that intersection of any two subgroups of a group $G$ is again a subgroup of $G$. (10)

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**DE–8038** **23**

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, MAY 2014.

OBJECT ORIENTED PROGRAMMING AND C++

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Write short notes on basic concepts of object oriented programming by explaining Data Abstraction, Dynamic binding and message passing.

   (b) What is scope resolution operator? Explain where and how it is used.

2. (a) What are the benefits of using oops programs? Explain.

   (b) What are manipulators? Explain.

3. (a) What are inline functions? Give an example and explain its use.

   (b) What are friend functions? What are the merits and demerits of using friend functions? Explain.

4. (a) What is constructor? How it is declared and defined? Explain with example.

   (b) Describe the access specifier with suitable example.

5. (a) Write a C++ program to generate a series of Fibonacci numbers using a copy constructor where the copy constructor is defined with in the class declaration itself.

   (b) Explain with an example how 'this' pointer is used in C++.

6. (a) What is operator overloading? Overload the unary operator '++' to the class 'counter'. It consists of one integer data member 'count' write a C++ program to accomplish above.

   (b) What are the uses of new and delete operator? Explain.

7. (a) What is polymorphism? Explain with suitable example.

   (b) Write a C++ program to demonstrate the function overloading.

8. (a) Write a program in C++ to demonstrate the use of multiple inheritance.

   (b) Under which aspects virtual function is useful? Explain with suitable example.

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**DE–8039** **24**

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (Computer Science) (Lateral) DEGREE EXAMINATION, MAY 2014.

SCIENTIFIC COMPUTING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain briefly Gauss Jordan method to solve simultaneous equations. (10)

   (b) Apply Gauss elimination method to find the solution of the following system. (10)

$$10x + y + z = 12$$
$$2x + 10y + z = 13$$
$$x + y + 5z = 7.$$

2. (a) Solve the following system of equations by Gauss-Jacobi method. (10)

$$x + y + 54z = 110$$
$$27x + 6y - z = 85$$
$$6x + 15y + 2z = 72.$$

   (b) Using Lagrange's formula, find $y(6)$ given (10)

| | | | | | |
|---|---|---|---|---|---|
| $x$ : | 2 | 5 | 7 | 10 | 12 |
| $y(x)$ : | 18 | 180 | 448 | 1210 | 2028 |

3. (a) Using Gauss interpolation formula, obtain $f(3.5)$ from the following table : (10)

| $x$ : | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| $f(x)$ : | 2.626 | 3.454 | 4.784 | 6.986 |

(b) Explain about Trapezoidal rule. (10)

4. (a) Evaluate $\int_{-3}^{3} x^4 dx$ using Simpson's ½ rule. (10)

(b) What is meant by a measure of dispersion? State the different methods of measuring it. (10)

5. (a) Calculate the standard deviation from the following :

14, 22, 9, 15, 20, 17, 12, 11. (10)

(b) Fit a parabola of the second degree to the data given below : (10)

| Year : | 1985 | 1986 | 1987 | 1988 | 1989 |
|---|---|---|---|---|---|
| Sales ('000) : | 16 | 18 | 19 | 20 | 24 |

6. (a) Two cards are drawn at random from a well shuffled pack of 52 cards. What is the probability that :

(i) Both are aces (ii) Both are red. (10)

(b) Explain about the Bayes theorem. (10)

7. (a) In a certain sample of 2000 families, 1400 families are consumers of tea. Out of 1800 Hindu Families, 1236 families consume tea. Use $\chi^2$ test and state whether there is any significant difference between consumption of tea among Hindu and Non-Hindu families. (10)

(b) A sample of 1000 students from Bombay University was taken and their average weight was found to be 112 Ibs with a standard deviation of 20 Ibs. Could the mean weight of students in the population be 120 pounds? (10)

8. The following data gives the number of units produced by four different workers using three different machines.

| | Machine Type | | |
|---|---|---|---|
| Workers | A | B | C |
| 1 | 20 | 28 | 26 |
| 2 | 22 | 30 | 32 |
| 3 | 26 | 32 | 10 |
| 4 | 32 | 24 | 24 |

Test whether :

(a) The four workers differ with respect to mean productivity.

(b) Whether the mean productivity is the same for different machine type. (20)

**DE–8040** **25**

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (Computer Science) (Lateral)
DEGREE EXAMINATION, MAY 2014.

APPLICATION PROGRAMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Write a short note on

   (i) Tool bar

   (ii) Dialog boxes. (10)

   (b) (i) How to work with multiple window? Explain.

   (ii) Write short notes on windows desktop. (10)

2. (a) Explain the various commands for formatting text in MS-Word. (10)

   (b) Explain the method of handling table in MS-Word. (10)

3. (a) What is mail merge? Explain the method of merging a form letter with data. (10)

   (b) Explain the various printing commands in MS-Word. (10)

4. (a) Explain the various functions in MS-Excel. (10)

   (b) How to formatting the data in MS-Excel? Explain. (10)

5. Explain the various types of charts available in MS-Excel with example. (20)

6. (a) What is template? How to use template in MS-Power point? Explain. (10)

   (b) Explain the text formatting and alignment in slide presentation. (10)

7. (a) Define primary key. Explain the procedure for creating a table in MS–Access. (10)

   (b) How to add, insert and delete a record in a database? Explain. (10)

8. (a) Explain the procedure for creating and printing reports. (10)

   (b) Explain the procedures for combining excel data and charts with word documents. (10)

**DE–8041** **31**

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc.(Lateral) DEGREE EXAMINATION, MAY 2014.

COMPUTER GRAPHICS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Briefly explain the various Input and Output devices. (10)

   (b) Discuss on the applications of computer graphics. (10)

2. (a) Define clipping. Explain Sutherland Hodgman algorithm for clipping. (10)

   (b) Explain point clipping with an algorithm. (10)

3. Explain the various types of three dimensional transformation with matrix representation. (20)

4. (a) Discuss on Z-buffer algorithms. (10)

   (b) Explain clipping process in a three dimensional plane. (10)

5. (a) Discuss the components of user Interface. (10)

   (b) Explain the usage of the user's model. (10)

6. (a) Explain Bresenham's algorithm for line drawing.(10)

   (b) Explain circle generation algorithm with example. (10)

7. (a) Discuss on viewing transformation. (10)

   (b) Define projection. Explain the various types of projection. (10)

8. (a) Explain in detail about command language. (10)

   (b) Write short note on Information display. (10)

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**DE–8042** **32**

DISTANCE EDUCATION

B.Sc.(Computer Science)/(B.Sc. Computer Science (Lateral))
DEGREE EXAMINATION, MAY 2014.

VISUAL PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss the features of windows graphical user interface concept. (10)

   (b) Narrate the hardware components and softwares that required to set up windows plat form. (10)

2. With proper procedure, explain the ways and means of selecting, moving and sizing windows. (20)

3. (a) Illustrate and explain the clipboard feature. (10)

   (b) Elucidate the importance of drawing tools and picture attributes. (10)

4. (a) Write about OLEDB and ODBC concepts. (12)

   (b) What is meant by event driven programming? Explain with an illustration. (8)

5. Describe the procedure of adding a menu to a form. Also write the procedure of creating submenus. (20)

6. (a) Write about the procedure of declaring arrays and dynamic arrays. (10)

   (b) With suitable illustrations, explain the ways of handling operators and operator precedence. (10)

7. With syntax and suitable illustrations explain the following looping concepts.

   (a) DO loop

   (b) For loop

   (c) While loop

   (d) If–else loop $(4 \times 5 = 20)$

8. (a) Give a note on MDI forms. Also discuss on arranging MDI child windows. (10)

   (b) Explain the procedure followed in arrays of forms. Also explain the procedure of passing forms to procedures. (10)

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**DE–8043** **33**

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (Computer Science) (Lateral)
DEGREE EXAMINATION, MAY 2014.

OPERATING SYSTEMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Write a Detailed note on Clustered and Handheld Systems. (10)

   (b) Discuss System and its various types in Detail. (10)

2. (a) Explain Message Passing in Detail. (10)

   (b) Write a detailed note on Process Scheduling. (10)

3. (a) List and explain the four circumstances under which the decisions for CPU scheduling takes place. (10)

   (b) Write a note on FCS and Priority Scheduling. (10)

4. (a) Define an Interrupt. Explain how will you handle an Interrupt if an Interrupt Occurs. (10)

   (b) Write a detailed note on Bankers Algorithm and Circular Wait. (10)

5. (a) What is Memory Management? Explain in detail about how to Manage Memory with Bitmaps. (10)

   (b) Define Paging. Explain various Techniques for Structuring the Page Table in Detail. (10)

6. (a) Explain in Detail about any two Page Replacement Algorithms. (10)

   (b) Write a note on interrupt handlers in detail. (10)

7. (a) Discuss in Detail about the various operations that can be performed on Files. (10)

   (b) Explain any two methods of Accessing Files. (10)

8. (a) What is a File? List out some File types with their Corresponding Extensions and the usage of that particular File type. (10)

   (b) Explain the various ways of improving the Reliability and performance of File Systems. (10)

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**DE–8044** **34**

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, MAY 2014.

INTERNET PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss about the seven layers of ISO OSI reference model. (10)

   (b) What is a network topology? Explain the different network topologies. (10)

2. (a) Write a short note on Internet address protocols. (10)

   (b) Explain the usages of FTP and Telnet. (10)

3. (a) Explain the term Domain Name System. Why it is used? (10)

   (b) What is e-mail? Explain its working with the help of an example. (10)

4. (a) What is E-Commerce? Describe the applications of E-Commerce. (10)

   (b) What are the various applications of the Internet? Explain. (10)

5. (a) What is an object-oriented programming? Why Java is known as platform independent language? (10)

   (b) Describe the type conversion and casting functions of Java. (10)

6. (a) List the basic data types used in Java. Explain with suitable example. (10)

   (b) Write a Java program to arrange the given set of numbers in an ascending order. (10)

7. (a) Explain about the single inheritance with an example. (10)

   (b) What is an applet? Explain the life cycle of an applet. (10)

8. (a) Write an applet to draw National flag with color.(10)

   (b) What is constructor? What are the different types of constructor? Explain. (10)

**DE–8045** **35**

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, MAY 2014

RDBMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$ marks)

1. (a) Define the term "DBMS". What are the advantages of DBMS? (10)

   (b) What is a database schema? Explain about the schema and subschema of the database. (10)

2. (a) Explain the terms primary key, candidate key and foreign key. Give an example for each. (10)

   (b) What is a relational model? Explain its structure. (10)

3. (a) Explain about the mapping operations of relational database. (10)

   (b) Explain about the data control facilities in relational database. (10)

4. (a) Write down the benefits of Oracle. (10)

   (b) What are the different types of processes in Oracle? Explain. (10)

5. (a) Discuss the logical structure of Oracle. (10)

   (b) Explain about the DML commands of SQL. (10)

6. (a) Illustrate with examples, how can you join multiple tables. (10)

   (b) What are the DDL commands in SQL? Explain with examples. (10)

7. (a) What is an index? Explain with an example. (10)

   (b) What is a privilege? Describe with examples. (10)

8. (a) What are the advantages of embedded query language? Give an example of an embedded SQL query. (10)

   (b) Explain how can you modify, rename and drop a table. (10)

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**DE–8799** **16**

DISTANCE EDUCATION

B.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2014.

COMPUTER LAB I — C AND DATA STRUCTURES LAB

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the students by lot system.

1. (a) Write a C program to find the factorial of a given number using function declaration.

   (b) Write a C program for linked list implementation of queue operations.

---------- Cut here ----------

2. (a) Write a C program to find simple and compound interests using function declaration.

   (b) Write a C program to evaluate the given mathematical expression using stack.

---------- Cut here ----------

3. (a) Write a C program to sort a set of elements using selection sort.

   (b) Write a C program to implement push and pop operations on stack.

4. (a) Write a C program to compute commission earned by a salesman according to the scheme given below :

| Sales Amount | Commission |
|---|---|
| Upto 10,000 | 5% |
| Above 10000 to 50000 | 10% |
| Above 50000 | 20% |

(b) Write a C program to implement insert and delete operations on linked list.

---------- Cut here ----------

5. (a) Write a C program to convert the given number into words for numbers 1 to 10.

(b) Write a C program for linked list implementation of queue operations.

---------- Cut here ----------

6. (a) Write a C program to sort a set of elements using insertion sort.

(b) Write a C program to implement insert and delete operations on queue using array.

**DE–8800** **26**

DISTANCE EDUCATION

B.Sc. (Computer Science)/ B.Sc.(CS)(Lateral) DEGREE EXAMINATION, MAY 2014.

LAB II — C++ PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the student by LOT system.

Each sub-division carries 50 marks.

1. (a) Write a C++ program to reverse the sentence and find the given sentence is palindrome or not.

   (b) Create a C++ class for a stock item abstract data type. It should have the attributes of stock levels (an integer) and unit price (float). Define the methods to returns the values of these two attributes and to set them using pointers. Add two more methods to allow stock receipts and issues updating the stock level as appropriate. Write a menu driven C++ program to solve the problem.

---------- Cut here ----------

2. (a) Using dynamic constructors, write a C++ program to concatenate two given strings.

   (b) Create a class int to overload all five integer arithmetic operators (+, –, *, / and %) so that they operate on objects of type int. If the result of any such arithmetic operation exceeds the normal range of integer from – 32768 to 32767 then print a warning and terminate the program. Write a program to test this class.

3. (a) Using pointers create a class and write a program to get the *n* names and display them in sorted order.

   (b) Write a C++ program to design a calculator with four arithmetic operations such as addition, subtraction, multiplication and division.

---------- Cut here ----------

4. (a) Write a menu driven C++ program to add and subtract given two matrices of order $m \times n$ defined in a class. Use operator overloading concept.

   (b) Create a class called employee that contains a name and an employee number. Include a member function called getdata( ) to get data from the user, another function called putdata ( ) to display the data. Write a main ( )program to exercise this class. It should create an array of type employee and then invite the user to input data for n employees.

---------- Cut here ----------

5. (a) Assume you want to generate a table of multiples of any given number. Write a program that allows the user to enter the number and then generates the table, formatting it into ten columns and 20 lines.

   (b) Write a program using friend function frifunc( ) which can act on the classes alpha and beta. Using constructors fix the values for alpha and beta.

---------- Cut here ----------

6. (a) Using operator overloading, write a C++ program to find the different and total length of given two various tubes specified in meters and centimeters.

   (b) Write a program that reads a group of numbers from the user and places them in array of type float. Once the numbers are sorted in the array, the program should average them and print the result. Use pointer notation whenever possible.

**DE–8801** **27**

DISTANCE EDUCATION

B.Sc. (CS)/ B.Sc.(CS)(Lateral) DEGREE EXAMINATION, MAY 2014.

Computer Lab – III

APPLICATIONS PROGRAMS LAB

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select Two questions one from Section – A and another from Section – B.

The questions are given to the candidates by lot system.

SECTION A – (50 marks)

1. Prepare a news report using two columns, insert a picture in the first column and make the text flow around it.

---------- Cut here ----------

2. Create a work sheet in MS-Excel with following columns : Employee number, Employee name, Designation, Basic pay, HRA, DA, LIC, PF, Gross pay, Net pay

   (a) Type data for empname, empno, designation, Basic pay, LIC, PF

   (b) Calculate HRA = 20% of Basic pay
   DA = 30% of Basic pay
   Gross pay = Basic pay + HRA + DA
   Net pay = Gross pay – (LIC + PF)

   (c) Draw the bar chart between employee name and net pay.

3. The following were the observations made in certain experiments for the values of $y$ and given the values of $x$.

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $x$ : | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| $y$ : | 10 | 30 | 45 | 25 | 15 | 28 | 40 | 32 | 15 | 35 |

Compute the following :

(a) Find mean, median and standard deviation

(b) Correlation coefficient between $x$ and $y$

(c) Draw the bar chart and shading cells

(d) Draw a chart in Excel and paste it on word document.

---------- Cut here ----------

SECTION B – (50 marks)

4. Create employee table with the fields EMP NO, NAME, AGE, SEX, STREET, CITY, PIN, SALARY.

(a) Add data for 20 employees

(b) Write a query to display all the male employees whose salary is between 1000 and 5000 and living in city "CHENNAI"

(c) Write a query to display all the female employees whose ages are in the range of 50-60

(d) Show all the records in the table for the city "TRICHY" by filtering.

5. Create a STUDENT table with the following fields REGNO, NAME, MARK 1, MARK 2, MARK 3. Create ADDRESS table with fields REGNO, STREET, CITY and PIN

   (a) Write a Query to display REGNO, NAME and total of all the three subject marks.

   (b) Write a query to display REGNO, NAME, STREET, CITY, PIN and total of all the three subject marks.

---------- Cut here ----------

6. Create inventory table with the fields ITEM-NO, NAME, QUANTITY-ON-HAND, REORDER-LEVEL

   (a) Create a form in custom format.

   (b) Create a query REORDER to show all the items.

   (c) Create a macro that executes REORDER query automatically.

**DE–8802** **36**

DISTANCE EDUCATION

B.Sc.(CS)/B.Sc.(CS) (Lateral) DEGREE EXAMINATION, MAY 2014.

COMPUTER LAB IV – VISUAL BASIC AND ORACLE LAB

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select and give ONE question to each candidate by lot system.

1. (a) Write and test a VB program to print 20 address labels in a two–column format for the address.
The Registrar,
University of Madras,
Chennai –600 006.

(b) Create a table BLOOD with the following fields: Donor number, name, date of birth, age, address, blood group and sex.
(i) Add 6 records
(ii) Display the information of donors in the age 20 to 25 using BETWEEN options.
(iii) Display the information of female donors with age 20 to 25 and having blood group A+.

---------- Cut here ----------

2. (a) Write a VB program to calculate simple interest or compound interest using three command buttons namely input, calculate and display.

(b) Create a table PERSONAL with the following fields. Serial number, name, address and phone
(i) Add 5 records
(ii) Display all names without duplicate
(iii) Display name, , city in which only beginning portion of name is known.

3. (a) Write and test a VB program to display the day of the given date and convert the given word from lower case to upper case.

(b) Create a table BIODATA with the following fields:

Name, age, sex, degree and city.

(i) Add 5 records

(ii) Display all records of male employees

(iii) Display all records having age <20 and city is Madurai.

---------- Cut here ----------

4. (a) Write and test a VB program to compute and print either the sum of odd numbers or even numbers at the user's choice using label, text and option buttons.

(b) Create a table RESULT with the following fileds:

Register number, name, mark 1, mark 2, mark 3, total and result.

(i) Add 5 records (Except total and result filed)

(ii) Fill total filed with the sum of mark 1, mark 2, mark 3.

(iii) Fill the result field with pass if total $\geq 150$ otherwise fail.

---------- Cut here ----------

5. (a) Write and test a VB program that allows the user choice among four arithmetic operations of addition, subtraction, multiplication and division with two given numbers.

(b) Create a table address with the following fields:

Name, street, city and pin

(i) Add 6 records

(ii) Add the field district

(iii) Fill all district field with Trichy.

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DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, MAY 2014.

COMPUTER LAB V JAVA PROGRAMMING LAB

(2007 onwards)

Time : Three hours Maximum : 100 marks

ONE question should selected by the examiner for each candidates and given to them by LOT system.

1. (a) Write a Java program to draw the various polygons using applet.

   (b) Write a program to read and write the data using random access file.

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2. (a) Write a Java program to implement producer /consumer problem using synchronization.

   (b) Write a Java program to display the contents of file along with a line number before each line.

---------- Cut here ----------

3. (a) Write a Java program to create multiple threads in Java.

   (b) Write a Java program that displays the number of characters, lines and words in a text file.

4. (a) Write a Java program to create a package.

   (b) Write a Java program that reads file name from the user and displays the contents of file.

---------- Cut here ----------

5. (a) Write a Java program which illustrate the implementation of multiple inheritance using interface.

   (b) Write a Java program to count the number of words and characters in a text.

---------- Cut here ----------

6. (a) Write a Java program to multiply the given matrices.

   (b) Write a Java program to manipulate labels, lists, text fields, text areas and panels.